

4. Gradient as a Functional

The gradient of a scalar valued function is denoted by its list of partial derivatives. Should it be considered a vector or a functional? Note that the gradient is usually used to express the change in the value of a function along a direction vector. This is a typical role of a functional.

Time Derivative Example

Consider a surface and a parametrization $\vec{x}(\vec{u})$

Also consider a scalar valued function $f(\vec{u})$ and a path $\vec{c}(t)$ in the parameter space. The function and path have compatible expressions on the surface as $f(\vec{x})=f(\vec{x}(\vec{u}))$ and $\vec{x}(\vec{c}(t))$.

The path velocity vector $\frac{d\vec{c}^k}{dt}$ at any point of the path in the parameter space

has its transformed version in the tangent space $\frac{\partial \vec{x}^i}{\partial u^k} \frac{d\vec{c}^k}{dt}$.

Note that the time rate of change of the function along the path should not change in either context. We use the chain rule to calculate the time derivative:

$$\frac{d}{dt} f(\vec{x}(\vec{c}(t))) = \left(\frac{\partial f}{\partial x^i} \right) \left(\frac{\partial x^i}{\partial u^k} \frac{d\vec{c}^k}{dt} \right) = \left(\frac{\partial f}{\partial x^i} \frac{\partial x^i}{\partial u^k} \right) \left(\frac{d\vec{c}^k}{dt} \right) = \frac{\partial f}{\partial u^k} \frac{d\vec{c}^k}{dt}$$

We can analyze the above middle equivalence in terms of functionals and vectors.

The left expression is the functional $\frac{\partial f}{\partial x^i}$ acting on the vector $\frac{\partial x^i}{\partial u^k} \frac{d\vec{c}^k}{dt}$ in the tangent space.

The right expression is the functional $\frac{\partial f}{\partial x^i} \frac{\partial x^i}{\partial u^k}$ acting on the vector $\frac{d\vec{c}^k}{dt}$ in the parameter space.

This example exhibits the role of transformation rules in conserving the result of a functional acting on a vector.

Later, we will see how to express the gradient as a vector.

Transformation of a Dot Product and g_{ij}

Consider a simple parameterization of a Cartesian space $\vec{x}(\vec{u})$ resulting from a change of variables (like polar coordinates). Consider two vectors in the parameter space ($\vec{u}$ and $\vec{v}$) and their images in the Cartesian space

($\vec{x}$ and $\vec{y}$), We have a way of calculating $\vec{x} \cdot \vec{y}$ in terms of $\vec{u}$ and $\vec{v}$:

$$\vec{x} \cdot \vec{y} = x^i \delta_{ij} y^j = \left(u^m \frac{\partial x^i}{\partial u^m} \right) \delta_{ij} \left(v^n \frac{\partial x^j}{\partial u^n} \right) = u^m g_{mn} v^n \text{ where } g_{mn} = \frac{\partial x^i}{\partial u^m} \delta_{ij} \frac{\partial x^j}{\partial u^n} \text{ is the}$$

array of all pairwise dot products of images of the the parameter space basis. In matrix terminology, this is also the transpose of the Jacobian matrix of $\vec{x}(\vec{u})$ multiplied by the original Jacobian matrix.

For polar coordinates it works out to $g_{ij} = \begin{bmatrix} 1 & 0 \\ 0 & r^2 \end{bmatrix}$

For a parameterized surface embedded in a Cartesian space, we can similarly pull back the dot product in the tangent planes and express it in terms of g_{ij} .

For the earth example and vectors in a tangent plane at a point identified by α and β :

$$g_{ij} = \begin{bmatrix} R^2 \cos^2 \beta & 0 \\ 0 & R^2 \end{bmatrix} \text{ (recall } \beta \text{ is latitude)}$$

Arc Length

The arc length of a curve on a surface embedded in a Euclidean space, can be approximated by summing the length of vectors v^i tangent to the curve. To this end consider a surface parametrized by $\vec{x}(\vec{u})$ and a path in the parameter space $\vec{c}(t)$. The path is carried to the surface as $\vec{x}(\vec{c}(t))$.

The arc length is approximated as $\sum \sqrt{\left(\frac{d}{dt} \vec{x}(\vec{c}(t)) \right) \cdot \left(\frac{d}{dt} \vec{x}(\vec{c}(t)) \right)} \Delta t$

The dot product that expresses vector length in the Cartesian space can be calculated in the parameter space using g_{ij} . The approximating sum can be pulled back to the parameter space which approximates the integral representing arc length:

$$\int \sqrt{\frac{dc^i}{dt} \frac{dc^j}{dt} g_{ij}} dt$$

Later we will discuss geodesics, that have the smallest path integral between its endpoints.

Note that the integrand expression specifying the arc length also can specify g_{ij} because of its symmetry properties.

$$\text{If arc length} = \int_{t_1}^{t_2} \sqrt{a\dot{x}^2 + 2b\dot{x}\dot{y} + d\dot{y}^2} dt \text{ then } g_{ij} = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$$

Surface Area (or Volume)

Consider a 2-d surface parametrized by $\vec{x}(\vec{u})$. A region on the surface has a corresponding region in the parameter space that can be divided into rectangles spanned by small vectors along the unit vectors $\Delta u^i \hat{u}_i$.

The spanning vectors of corresponding parallelograms in the embedding space are given by $\Delta u^k \frac{\partial \vec{x}^i}{\partial u^k}$ in the tangent space to the surface. The summed areas of these parallelograms approximate the area. The parallelograms in the embedding space do not bend with the surface. They are in the tangent planes, pinned to the surface at the tangent plane origin, like roof shingles.

We don't calculate the area using the vector cross product:

$$A = \left| \left(\Delta u^1 \frac{\partial \vec{x}^k}{\partial u^1} \right) \times \left(\Delta u^2 \frac{\partial \vec{x}^k}{\partial u^2} \right) \right| \text{ because it doesn't generalize to higher dimensions.}$$

Previously, we saw that the area^2 of a parallelogram embedded in a space of higher dimension is given by the determinant of the matrix of pairwise dot products of the images of the parameter space basis. In this context, the area squared of each parallelogram is given by the determinant of the matrix whose

$$mn^{\text{th}} \text{ element is given by the dot product } \left(\Delta u^m \frac{\partial \vec{x}^k}{\partial u^m} \right) \cdot \left(\Delta u^n \frac{\partial \vec{x}^k}{\partial u^n} \right).$$

So, we use:

$$\text{Area}^2 = \left| \Delta u^m \frac{\partial \vec{x}^k}{\partial u^m} \Delta u^n \frac{\partial \vec{x}^k}{\partial u^n} \right| = \left| \frac{\partial \vec{x}^k}{\partial u^m} \frac{\partial \vec{x}^k}{\partial u^n} \right| (\Delta u^1)^2 (\Delta u^2)^2 = |g_{mn}| (\Delta u^1)^2 (\Delta u^2)^2$$

In the above expression, the Δu^i 's can be factored out of the determinant, because they are constant multiples of a row or column.

So that the total area is calculated as:

$$\sum \sqrt{|g_{mn}|} \Delta u^1 \Delta u^2 \rightarrow \int \sqrt{|g_{mn}|} du^1 du^2$$

For polar coordinates $|g_{mn}|=r^2$ and area is calculated with the familiar $\int r dr d\theta$

Changing Parameterizations

Transition rules for vectors and functionals between parameter spaces in overlapping regions are the same as transition rules between a parameter space and a surface.

Consider two parameterizations of a surface in an embedding space with variables $\vec{z}$. Let the two parameterizations over some overlapping region be given by $\vec{z}(\vec{x})$ and $\vec{z}(\vec{y})$ where the parameter spaces have variables $\vec{x}$ and $\vec{y}$. Then the parameter spaces can parametrize each other by following a point between the surface and the parametrization spaces. We can find a transition function like $\vec{y}(\vec{x})$. This is like identifying the same point on two different geographical map projections.

Consider a vector v^i or a covector a_i in the parameter space with $\vec{y}$ variables, and corresponding vector v^k or covector a_k in the space with $\vec{x}$ variables. They must satisfy the dual transformation rules:

$v^i = v^k \frac{\partial y^i}{\partial x^k}$ and $a_k = a_i \frac{\partial y^i}{\partial x^k}$ where the derivative terms change from point to point.

Note that, these transformation rules do not need to refer to the embedded surface as long as the transition functions like $\vec{y}(\vec{x})$ are available.

The transformation rules could be written in a form where the derivatives of the inverse transition function is used for transforming functionals:

$$v^i = v^k \frac{\partial y^i}{\partial x^k} \text{ and } a_i = a_k \frac{\partial x^k}{\partial y^i}$$

We don't use this form as often when the derivatives in one direction have simpler expressions than the other. There are reasons to use the other form when studying induced functions.

If required, we can find $\frac{\partial x^k}{\partial y^i}$ by inverting the matrix $\frac{\partial y^i}{\partial x^k}$.